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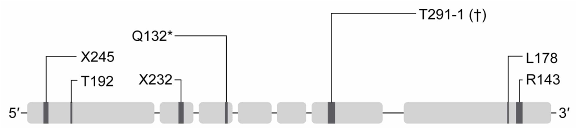

Figure 1 Loss-of-function mutations observed in the *ANGPTL4* gene of participants without CAD (Note: * indicates mutation resulting in a truncated protein; † indicates substitution of a cytosine for its complementary base.)

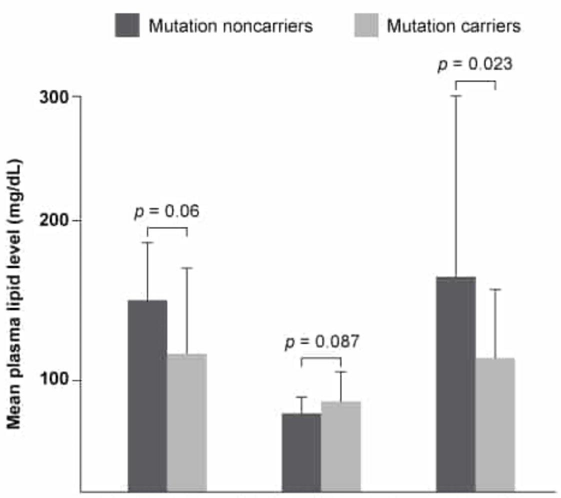

Figure 2 Mean plasma lipid levels for *ANGPTL4* mutation carriers and noncarriers. (Note: LDL = low-density lipoprotein; HDL = high-density lipoprotein; error bars = standard deviation.)

To further analyze the effect of *ANGPTL4* on blood lipid levels, *ANGPTL4* knockout mice and mice that constitutively express *ANGPTL4* were created. Researchers hypothesized that two of the four biological profiles shown in Table 1 would be observed in these mice.

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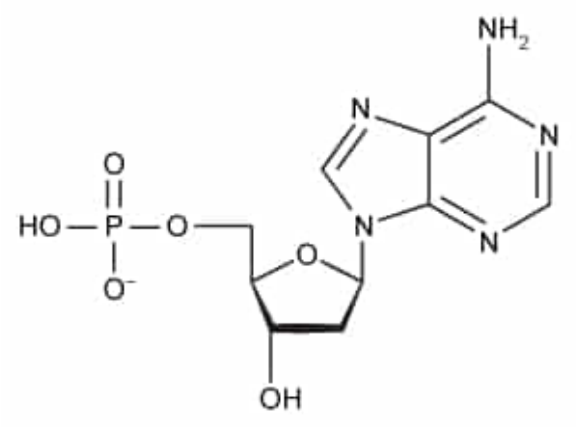
Profile 1	None	None	Decreased	Increased
Profile 2	Increased	Increased	Increased	Decreased
Profile 3	None	None	Increased	Decreased
Profile 4	Increased	Increased	Decreased	Increased

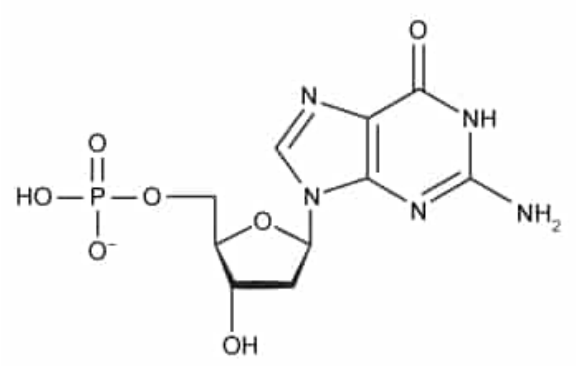
Based on Figure 1, substitution of cytosine for which of the following nucleotides would cause the T291-1 loss-of-function mutation?

A.



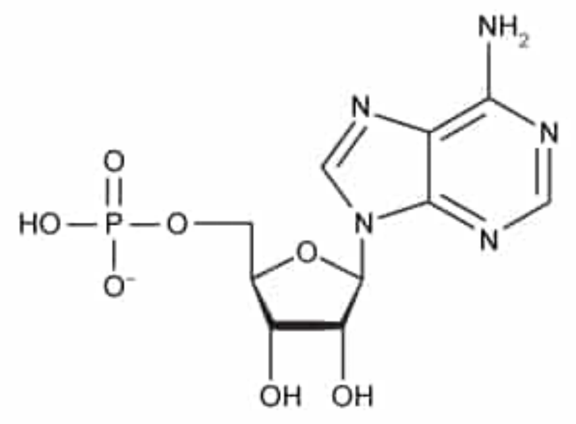
[16%]

B.



[61%]

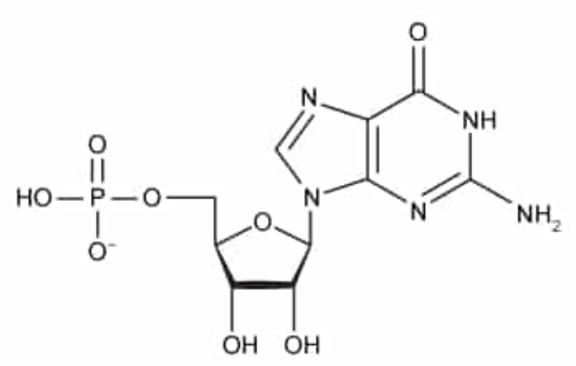
C.



[8%]

D.

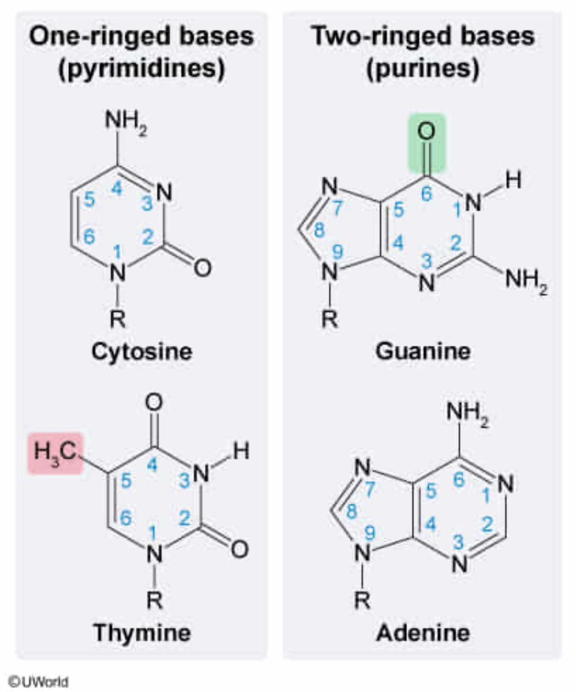
微信Mai_ange



[13%]

Correct
61% answered correctly

Explanation:



Deoxyribonucleotides, the building blocks of DNA, are each composed of a phosphate, a deoxyribose sugar, and one of four nitrogenous bases (A, T, C, or G). Individual nucleotides can be identified by analyzing the structure of their attached base:

- Cytosine (C) and thymine (T) are **one-ringed** bases known as pyrimidines. Thymine contains a **methyl group (–CH_3)** attached to the carbon atom in the fifth position, a feature that helps distinguish it from cytosine.
- Guanine (G) and adenine (A) are **two-ringed** bases known as **purines**. Guanine has a **carbonyl group (–C=O)** at the sixth position that helps differentiate it from adenine.

In double-stranded genetic material, **complementary base pairing** is the bonding of a particular nucleotide with only one other type of nucleotide. Such base pairing is determined by the alignment of the hydrogen bond donors and acceptors of each nucleotide's nitrogenous base. Cytosine and guanine always base pair via three hydrogen bonds. The caption for Figure 1 states that the T291-1 mutation in *ANGPTL4* results after substituting a **cytosine** with its **complementary base** (**guanine**), which is depicted in Choice B.

(Choice A) The nucleotide here contains an adenine base, which is complementary to thymine (*not* cytosine). In DNA, adenine always base pairs with thymine via two hydrogen bonds. In RNA, adenine base pairs with uracil (U).

(Choices C and D) These nucleotides contain a ribose sugar with a hydroxyl group (–OH) on the 2′ *and* 3′ carbon atoms. Therefore, these are ribonucleotides found in RNA (*not* deoxyribonucleotides found in DNA).

Educational objective:
Deoxyribonucleotides are composed of a phosphate, a deoxyribose sugar, and a nitrogenous base (A, T, C, or G). Cytosine and thymine are pyrimidines (single-ringed bases) whereas guanine and adenine are purines (double-ringed bases). In double-stranded DNA, cytosine always aligns with guanine (and vice versa), and adenine always aligns with thymine (and vice versa) via complementary base pairing.

Time spent:0 sec
Subject:
Biochemistry

QID:401541
System:
Carbohydrates, Nucleotides, and Lipids

Abnormal blood lipid levels play a significant role in the development of coronary artery disease (CAD). The discovery of genomic variations that confer a protective effect against CAD has prompted investigations into the biological impact of these protective alleles. Researchers have observed that specific single-base variations in the *ANGPTL4* gene are common among patients with a low occurrence of CAD. *ANGPTL4* is located on chromosome 19 and codes for angiopoietin-like 4, a multifunctional glycosylated protein.

One of the functions of angiopoietin-like 4 is the regulation of lipid metabolism. Specifically, angiopoietin-like 4 inhibits the activity of the enzyme lipoprotein lipase (LPL). Generally, active LPL is a homodimer consisting of two individual subunits that are each approximately 450 residues in length and weigh 50-kDa. LPL functions on the surface of vascular endothelial cells to degrade the triglycerides of chylomicrons and very-low-density lipoproteins (VLDLs) in the blood. Elevated triglyceride levels have been correlated with increased CAD risk.

To identify common loss-of-function mutations in *ANGPTL4*, DNA samples from 43,000 individuals with CAD and 78,000 individuals without CAD were genotyped. A detailed analysis of *ANGPTL4* genomic variants was performed, and the results of individuals without CAD were mapped in Figure 1. The lipid profiles of *ANGPTL4* mutation noncarriers were also compared to those of mutation carriers (Figure 2).

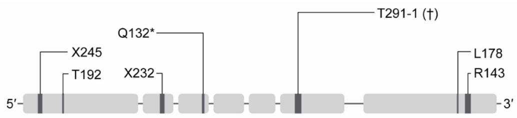


Figure 1 Loss-of-function mutations observed in the *ANGPTL4* gene of participants without CAD (Note: * indicates mutation resulting in a truncated protein; † indicates substitution of a cytosine for its complementary base.)

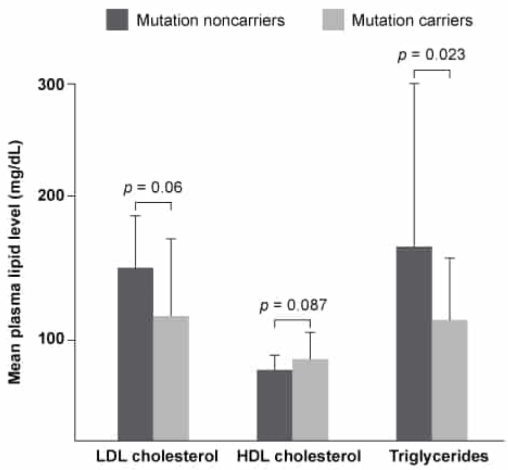


Figure 2 Mean plasma lipid levels for *ANGPTL4* mutation carriers and noncarriers. (Note: LDL = low-density lipoprotein; HDL = high-density lipoprotein; error bars = standard deviation.)

To further analyze the effect of *ANGPTL4* on blood lipid levels, *ANGPTL4* knockout mice and mice that constitutively express *ANGPTL4* were created. Researchers hypothesized that two of the four biological profiles shown in Table 1 would be observed in these mice.

Table 1 Predicted Biological Profiles of *ANGPTL4* Mouse Models

	ANGPTL4 mRNA	Angiopoietin-like 4 activity	Lipoprotein lipase activity	Serum triglyceride levels
Profile 1	None	None	Decreased	Increased

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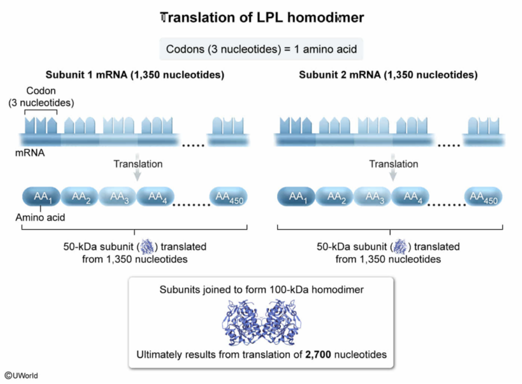
Profile 3	None	None	Increased	Decreased
Profile 4	Increased	Increased	Decreased	Increased

A scientist studying the sequence of the LPL homodimer would most likely expect that:

- A. ribosomes must translate a total of 1,350 mRNA nucleotides into amino acids to form one 100-kDa LPL homodimer. [16%]
- B. ribosomes must translate a total of 2,700 mRNA nucleotides into amino acids to form one 100-kDa LPL homodimer. [57%]
- C. ribosomes must translate a total of 1,350 mRNA nucleotides into amino acids to form one 50-kDa LPL homodimer. [20%]
- D. ribosomes must translate a total of 2,700 mRNA nucleotides into amino acids to form one 50-kDa LPL homodimer. [5%]

Correct
57% answered correctly

Explanation:



A dimer is a protein generally composed of two noncovalently bound polypeptide chains (also called subunits, or monomers). A **homodimer** is a protein in which the polypeptide chains of the two monomers have the same sequence (number, order, and type) of amino acids (AAs). According to the passage, active LPL is generally a homodimer consisting of two individual subunits, which each are approximately 450 amino acid residues in length and weigh 50-kDa. Accordingly, the **LPL homodimer** (consisting of *two* subunits) is made up of approximately **900 total amino acid residues** and weighs around **100-kDa**.

In this scenario, the scientist is calculating the *number of mRNA nucleotides* that ribosomes must translate into amino acids to form the 100-kDa LPL homodimer. Specifically, LPL mRNA must be translated *twice* to form each subunit.

First, the total number of mRNA nucleotides that must be translated by ribosomes to form one LPL subunit can be calculated as follows:

Number of AAs in a single LPL subunit × 3 nucleotides per codon = number of nucleotides in mRNA sequence

450 AAs in LPL subunit × 3 nucleotides = **1,350 nucleotides**

Because the mRNA must be **translated twice** to produce the **two subunits** of the LPL homodimer, the *total* number of mRNA nucleotides translated by ribosomes to form the homodimer can then be calculated:

Number of nucleotides translated to form a single subunit × 2 = total number of nucleotides translated to form homodimer

1,350 nucleotides × 2 = **2,700 total nucleotides**

Therefore, ribosomes must translate approximately **2,700 total nucleotides** to form one **100-kDa** LPL homodimer.

(Choices A, C, and D) Compared to the LPL homodimer, the *individual LPL subunits* are half the weight and are encoded by half the number of mRNA nucleotides. As a result, 1,350 mRNA nucleotides must be translated to form each 50-kDa LPL *subunit*.

Educational objective:
The total number of nucleotides in an mRNA molecule can be calculated by multiplying the number of amino acids in the protein by the number of nucleotides in a codon (ie, three).

Time spent:0 sec
Subject:
Biochemistry

QID:401542
System:
Carbohydrates, Nucleotides, and Lipids

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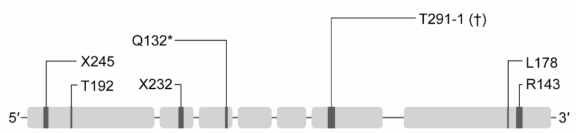
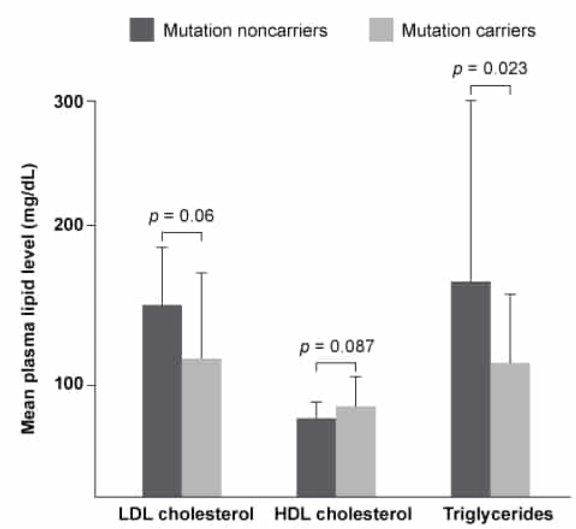


Figure 1 Loss-of-function mutations observed in the *ANGPTL4* gene of participants without CAD (Note: * indicates mutation resulting in a truncated protein; † indicates substitution of a cytosine for its complementary base.)



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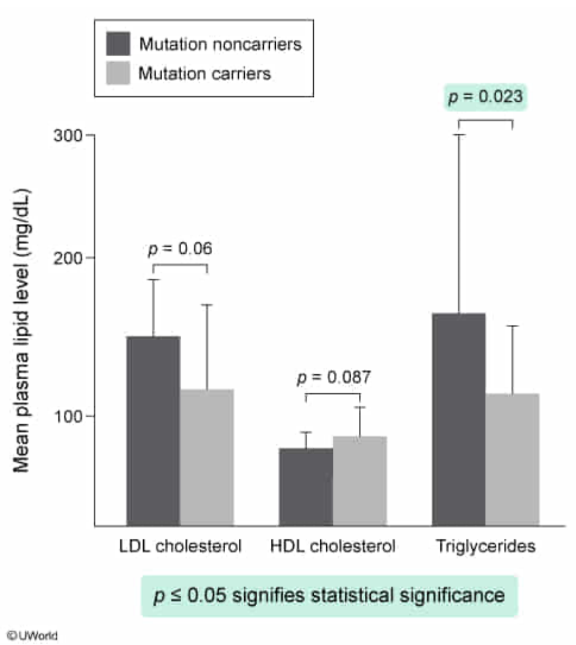
Based on Figure 2, *ANGPTL4* mutations were found to decrease CAD risk by significantly altering which of the following?

I. LDL cholesterol levels
II. HDL cholesterol levels
III. Triglyceride levels

- A. I only [4%]
- B. III only [85%]
- C. I and II only [5%]
- D. II and III only [4%]

Correct
86% answered correctly

Explanation:



The ***p*-value** is widely used to determine the significance of experimental results. A *p*-value is formally defined as the **probability** of **observing a given result** due to **chance**

alone, assuming that the null hypothesis (status quo assumption that there is no actual difference between the experimental and control groups) is true.

The *p*-value ranges from 0 to 1 and is generally interpreted as follows:

- $p > 0.05$ signifies a greater than 5% probability that the observed result is due to chance alone, *not* due to an actual association between the variables under study. Values in this range are generally not considered statistically significant.
- $\boldsymbol{p \leq 0.05}$ signifies a 5% or lower probability that the observed result is due to chance alone. Values in this range are typically considered **statistically significant**.

The plasma triglyceride level in mutation carriers is significantly lower than the plasma triglyceride level in mutation noncarriers ($p = 0.023$) **(Number III)**.

(Number I) LDL cholesterol has a *p*-value of 0.06, which is *not* considered statistically significant.

(Number II) HDL cholesterol has a *p*-value of 0.087, which is *not* considered statistically significant.

Educational objective:
The *p*-value is the probability of observing a result due to chance alone, assuming that the null hypothesis is true. A value of $p \leq 0.05$ is generally considered statistically significant whereas $p > 0.05$ is not considered statistically significant.

Time spent:0 sec	**QID**:401543
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

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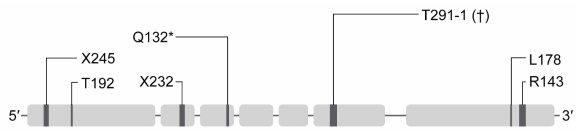

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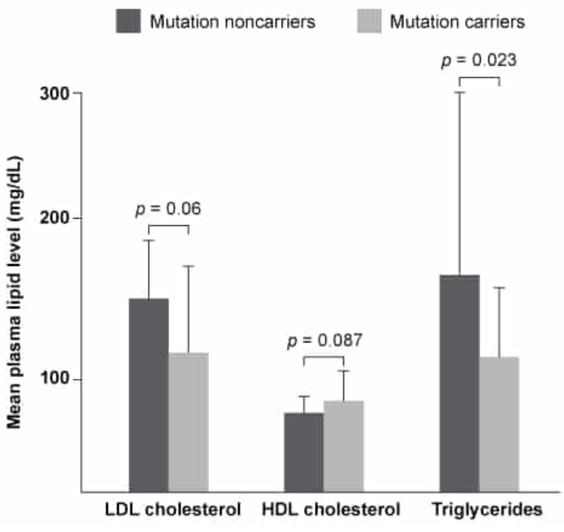

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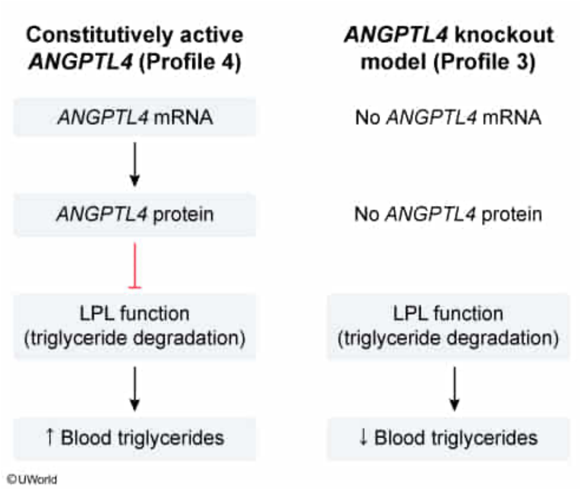
Profile 3	None	None	Increased	Decreased
Profile 4	Increased	Increased	Decreased	Increased

The knockout mouse model and the constitutively active mouse model of *ANGPTL4* are best represented by:

- A. Profile 1 and Profile 2, respectively. [27%]
- B. Profile 1 and Profile 4, respectively. [8%]
- C. Profile 3 and Profile 4, respectively. [56%]
- D. Profile 3 and Profile 2, respectively. [7%]

Correct
57% answered correctly

Explanation:



Per the passage, *ANGPTL4* codes for angiopoietin-like 4, a protein that inhibits the action of LPL. The passage also states that LPL catalyzes the degradation of triglycerides (lipid molecules) in the blood. Consequently, expression of the wild-type *ANGPTL4* gene, whose protein product inhibits LPL activity, leads to higher levels of triglycerides in the blood and therefore increases the risk for CAD.

Based on Table 1, Profile 3 represents the **knockout** mouse model in which the gene of interest (*ANGPTL4*) has been inactivated. This results in **absent *ANGPTL4*** mRNA and angiopoietin-like 4 protein activity. Consequently, **LPL activity** is **increased** due to the lack of inhibition, which causes *decreased* levels of blood triglycerides (as these are degraded by LPL).

Profile 4 represents the **constitutively active** mouse model in which *ANGPTL4* is transcribed at a relatively constant rate regardless of current cell conditions. This results in **increased *ANGPTL4*** mRNA and angiopoietin-like 4 protein activity. In turn, the angiopoietin-like 4 protein **inhibits LPL activity**, which leads to *increased* triglycerides in the blood.

(Choices A, B, and D) A knockout model suggests that *ANGPTL4* gene expression and consequently its mRNA are absent; this condition eliminates Profile 1 because loss of ANGPTL4 expression leads to increased (not decreased) LPL activity. In the constitutively active mouse model, *ANGPTL4* gene mRNA levels are increased due to constant *ANGPTL4* expression; this leads to decreased (*not* increased) activity of LPL, eliminating Profile 2 as an option.

Educational objective:
A constitutively active gene is transcribed at a relatively constant rate regardless of current cell conditions. In a knockout model, an existing gene is replaced or disrupted, which leads to an absence of the protein product.

Time spent:0 sec	**QID**:401544
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids